Subset Sums in a Fourfold Sumset

An integer prefix inverse and the audited bound $N \le 93$

Sum-of-Subset research project

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Abstract

We continue the independently audited chain from $N \le 94$ to $N \le 93$. A new exact integer inverse combines the threshold-17 non-progression lemma with a dichotomy on large excess and non-progression prefixes. We use small-hole witnesses from the first admissible parameter, rather than waiting until the hole count reaches 15. At cardinality 94 this reduces the positive-low-hole covering cap from 266 to 242, below the three-empty threshold 258. The low-hole root universe shrinks by 20,264 integer systems. The high-hole proof retains the newly possible two-empty family and groups 3,178 states into 69 common models while retaining their hole variable. Fresh exact certificates close the complete endpoint; archived endpoint certificates are not used. The stronger archived bound 84 has not yet received a complete independent replay, so this is progress in the audited chain, not a numerical improvement over that archive.

1 Result and inherited foundation

For a finite real set A, let $\mathrm{FS}(A) = \{\sum_{x\in X} x : X \subseteq A\}$ include zero, and let $4A = A + A + A + A$ permit repeated summands. Let N be the maximum feasible size under $\mathrm{FS}(A) \subseteq 4A$.

Theorem 1.1. *Every feasible set has at most 93 elements. Hence* $71 \le N \le 93$.

The lower bound is still $[-31, 31] \cap \mathbb{Z} \cup \{\pm 93, \pm 155, \pm 217, \pm 713\}$. The upper bound follows from the audited chain $115 \to 103 \to 100 \to 99 \to 98 \to 97 \to 96 \to 95 \to 94$ and the exclusion of size 94 below. We inherit the corrected peeling, integralization, small-increment inverse, simultaneous inverse, effective-hole, covering, mass, span, and sharp four-extreme lemmas from [3, 2, 1]. Their hypotheses are checked anew. No feasibility under arbitrary deletion is assumed.

For a positive r-set X, put $T_r = r(r+1)/2$ and

$$\mu(X) = |\mathrm{FS}(X)| - 1 - T_r, \quad \delta_0(X) = |\{x \in X : 2x \notin R(X \cup \{0\})\}|, \quad \rho(X) = \mu(X) + 2 - 2\delta_0(X),$$

where R consists of distinct-element two-sums. Increasing prefixes X_i have nonnegative integer increments $y_i = |\mathrm{FS}(X_i)| - |\mathrm{FS}(X_{i-1})| - i$, with

$$\mu(X_r) = \sum_i y_i, \qquad \rho(X_i) \ge \rho(X_{i-1}) + y_i - 2. \tag{1}$$

The previous report proves $\rho(X) \ge r - 5$ for $r \ge 17$ and non-progression $\mathrm{FS}(X)$, with sharp constant 5. Its 2,723 finite base cases are recomputed by the new verifier, without modifying that lemma or its independent checker.

2 An exact non-progression prefix inverse

The inherited inverse, for $2 \le \beta \le 10$, is

$$L_\beta(r, u) = \min\{(\beta-1)(r-14), (\beta-1)q + \min(s, \beta-1)\}, \quad \max(0, u - 4\beta + 2) = (\beta+1)q + s,$$

where $0 \le s \le \beta$. For a non-progression set of size $r \ge 17$ with $\mu \ge u$, define the valid old floor

$$V(r,u) = \max\{r-5,\, u+2-2r, L_2(r,u), \dots, L_{10}(r,u)\}. \tag{2}$$

The number 14 inside L_β is retained: substituting 5 there would not follow from the sharper non-progression theorem.

For integers $v \ge 0$ and $k \ge 1$, write $v = (k+2)q + s$, $0 \le s < k+2$, and define

$$J_k(v) = kq + \min(s,k). \tag{3}$$

This is the least nonnegative integer t with $t + 2\lfloor t/k \rfloor \ge v$. Indeed, on a block $t = kq + h$ with $0 \le h < k$, the left side is $(k+2)q + h$; the two skipped values are first reached at the next multiple of k.

Theorem 2.1 (Prefix inverse)**.** *Let $r \ge 17$, let $\mathrm{FS}(X)$ be non-progression, and suppose $\mu(X) \ge u$. For $17 \le L \le r$ and $3 \le \beta \le \min(10, L-4)$, put*

$$\begin{aligned} R_0 &= r-5, & M &= T_L - r,\\ A &= R_0 + (\beta-2)(r-L), & C &= V(r, M+1),\\ B &= R_0 + J_{\beta-2}\big(\max(0, u - R_0 - 4\beta + 2)\big). \end{aligned}$$

Then $\rho(X) \ge \min(A,B,C)$. Consequently

$$F(r,u) = \max\left(V(r,u),\ \max_{L,\beta} \min(A,B,C)\right) \tag{4}$$

is a valid lower bound for $\rho(X)$.

Proof. If $\mu \ge M+1$, use (2) to obtain $\rho \ge C$. Otherwise every prefix of size at least L has non-progression support. To see this, if a prefix at index $i \ge L$ were a progression, let $k > i$ be the first subsequent non-progression index. Its predecessor support and new translate must be disjoint: overlap puts the translating element on the same step and their union is another progression. It follows that

$$y_k = |\,\mathrm{FS}(X_{k-1})| - k \ge 1 + T_{k-1} - k \ge 1 + T_L - r = M+1,$$

contradicting $\mu = \sum y_i \le M$.

If every $y_i > \beta$ for $L < i \le r$, the floor at X_L and (1) give $\rho \ge L - 5 + (\beta-1)(r-L) = A$. Otherwise take the last $i > L$ with $y_i \le \beta$. The inverse theorem applies because $i \ge L+1 \ge \beta+5$. Up to scaling, X_i is an interval with at most β deletions, retaining its maximum. It has at most 2β diagonal defects. Here the elementary endpoint-band argument is sufficient: for $q \ge 2$ deletions and $i > q$, every defect lies in $[1,q]$ or $[i, i+q]$, and not all q low elements can be present and defective, since defectivity of 1 excludes 2. Thus there are at most $(q-1) + (q+1) = 2q$ defects. For $q = 1$ there are at most two, and for $q = 0$ only the maximum.

Write $t = r - i$. Each later step adds at most one defect and raises ρ by at least $\beta - 1$, so

$$\delta_0(X) \le 2\beta + t, \qquad \rho(X) \ge i - 5 + (\beta-1)t = R_0 + (\beta-2)t.$$

With $s = \rho - R_0 \ge 0$, the identity for μ yields

$$u - R_0 - 4\beta + 2 \le s + 2\left\lfloor \frac{s}{\beta-2} \right\rfloor.$$

The exact inverse (3) gives $\rho \ge B$. These cases exhaust all configurations and prove the theorem. □

The archive [4] contains a related prefix argument starting at 21 with a linear ceiling. The present proof starts at 17, permits $\beta \le L-4$ by searching strictly after L, and preserves the integer staircase rather than replacing the floor by a real quotient. For example, $(r,u) = (38,165)$ gives $V = 105$ but $F = 111$, witnessed by $(L,\beta,M,A,B,C) = (20,8,172,141,111,112)$. The checker verifies the exact inverse at 5,409 parameter pairs $1 \le k \le 9$, $0 \le v \le 600$, including minimality. This is a check of the finite formula used here; the proof of the theorem is the unbounded argument above.

3 The size-94 lattice gate

A size-94 counterexample contains zero, since adjoining zero otherwise would give a size-95 counterexample. After integralization, remove four original extremes on each side to get D, and peel twice to $D \supset D_1 \supset E = P \cup \{0\} \cup (-Q)$. Use

$$d = 86, \quad e = 78, \quad m = a + b = 77, \quad I = (a - 38)(a - 39),$$
$$\Lambda = 1 + T_a + T_b = 1522 + I, \quad \Omega = 467 - I.$$

The inherited identity is

$$\rho(P) + \rho(Q) + 2\vartheta + c + \mathcal{T} = \Omega, \qquad c = 2|2D| + 18d + 85 - |\operatorname{FS}(D)|,$$

with nonnegative terms, and $\mathcal{T} = \kappa + \ell_1 + \ell_2 + g$. The deficient-sign test is $1 + T_{78} > 24 \cdot 94 + 77$. The paired bootstrap gives $q^2 \le 2477$ with odd q, so $q \le 49$ and $a, b \ge 14$. Then $\Omega \ge 0$ leaves $17 \le a \le 60$. The simultaneous inverse condition holds for all 44 orientations:

$$T_{\min(a,b)} - 2 - \Omega \ge 38 > 0.$$

It supplies a common core lattice, with holes $z = h_P + h_Q - \kappa$, $0 \le z \le \min(h_P, h_Q) \le \kappa$ and $h_X < |\operatorname{FS}(X)| - 1$.

Write $\rho = \rho(P) + \rho(Q)$, $\mu = \mu(P) + \mu(Q)$ and $C_0 = 2m + 4e + 14 - \Lambda = -1042 - I$. The inherited inner comparisons are

$$|\operatorname{FS}(D_1)| - 2|\operatorname{FS}(E)| \le \min\{C_0 + \Omega + \rho - \mu + 2z,\ C_0 + 3\Omega - \rho - \mu\}.$$

For $z \le 14$, $\rho \le \mu$ makes the first bound at most $-547 - 2I$. For $z \ge 15$, both supports are non-progressions. The size-17 theorem already applies to the cores, without assuming the outer sets are on the lattice, giving $\rho + \mu \ge (m - 10) + (m - 6) = 138$. Whenever this is insufficient, use the effective-hole witness at $\beta = 7$ separately for each side. The necessary rows are

r	32	33	34	35	36	37	38
$C_7(r)$	83	89	96	102	109	110	116
r	39	40	41	42	43	44	45
$C_7(r)$	122	128	129	135	142	147	149

They give $\mu(X) \ge r + C_7(r) + 1$. With $\rho \ge m - 10$, the final high-hole comparison is at most -3. The outer margin is

$$\Lambda - (4d - 10 + \Omega) = 721 + 2I > 0.$$

Thus an off-lattice element would produce disjoint support translates, contradicting the relevant strict factor-two comparison. Both D_1 and D are on the core lattice. Only now use hole propagation and the sharp four-extreme theorem to obtain

$$w \le 4z, \qquad \mathcal{T} \ge \kappa + 162 - 3z, \qquad \mathcal{T} \ge \max(54, z, 162 - 2z). \tag{5}$$

The sharp non-progression theorem is valid for real positive sets and is independent of (5); this order of proof is not circular.

4 Hole-sensitive pruning and coupled groups

We use the effective-hole theorem from the foundation in its exact form. Given positive integers r, β and $C \ge 0$, let K be the least integer ≥ 3 with $(K + 1)(K - 2)/2 > C + 1$, and put

$$D_0 = C + K, \quad E_0 = D_0 + 1,$$
$$M_0 = \max\left(2\beta, 1 + \left\lfloor \frac{1 + \sqrt{9 + 8(E_0 + 6\beta)}}{2} \right\rfloor\right), \quad J = \max(K + 1, 2\beta + 5, M_0).$$

A witness is accepted only when all three integer inequalities hold:

$$T_{M_0} - 6\beta > M_0 + 1 + E_0, \qquad \beta(r - J + 1) > D_0, \qquad T_{J-1} - 1 > D_0.$$

If $2\beta < h_X < |\,\mathrm{FS}(X)| - 1$, it then gives $\mu(X) \geq r + C + 1$. The search uses $17 \leq r \leq 60$, $1 \leq \beta \leq 20$, $0 \leq C \leq 600$; only valid witnesses are used, and no optimality of these limits is claimed. Every chosen witness is checked again with exact integers.

For $z > 0$, let $U(r, z)$ be the maximum of $r - 3$ and all bounds $r + C + 1$ from accepted witnesses with $2\beta < z$. For $z = 1, 2$ only $r - 3$ is needed. Set $R(r, z) = F(r, U(r, z))$. The common covering cap is

$$c \leq 467 - I - R(a, z) - R(b, z) - \max(54, z, 162 - 2z). \tag{6}$$

For $z = 0$ use zero side penalties, obtaining $c \leq 305 - I$. All positive-hole states are enumerated within $1 \leq z \leq \Omega$; negative caps are discarded. The finite scalar audit gives

$$1 \leq z \leq 14 \implies c \leq 242 - I < 3d.$$

For $z = 1, 2$ use $R(a, z) + R(b, z) \geq 67$. For $z = 3, 4$, only the $\beta = 1$ witnesses are needed: (4) gives a combined floor at least 80 over all 44 orientations. For $5 \leq z \leq 14$, the $\beta = 2$ witnesses alone give at least 103. Thus

$$c + I \leq \max\{305 - 67 + 4,\ 305 - 80 + 8,\ 305 - 103 + 28\} = 242.$$

All 88 pairs of small-hole witnesses and inverse parameters are saved in the scalar audit. This step uses hole information before any transport. The former coarse estimate would only give $c \leq 266 - I$.

Let j be the number of empty covering units and $\delta = c - 86j$. The complete low-hole root ranges are

Regime	Z_0	Z_1	Z_2	Z_3	P_0	P_1	P_2
a	22–55	24–53	27–50	32–45	23–54	26–51	31–46
Integer roots	136	2,100	13,872	41,888	128	1,820	9,248

Here Z means zero holes and P means positive low holes. The old cap 266 would retain 31,460 positive roots, including 17,952 three-empty roots. The new positive count is 11,196: 20,264 fewer roots before geometric solving. With the unchanged 57,996 zero-hole roots, the complete low-hole universe has 69,192 integer models.

For $z \geq 15$, 3,896 oriented scalar states survive and $c \leq 184 < 3d$. The first pair's actual contributor fixes the direction in the inherited active-center estimate; reflection cannot be chosen to maximize it. There remain 1,350 states for $j = 0$, 1,746 for $j = 1$, and 82 for $j = 2$. For each fixed $(a, j, U(a, z), U(b, z))$, retain the actual hole count as one common variable z, bounded by the smallest and largest values in the group. The side excess floors are constant throughout this group. Write $E_* = 467 - I - F(a, U(a, z)) - F(b, U(b, z))$. Instead of independently maximizing the hole count and the cap, keep all three coupled rows

$$\delta + z \leq E_* - 86j, \qquad \delta \leq E_* - 54 - 86j, \qquad \delta - 2z \leq E_* - 162 - 86j.$$

They are exactly the three alternatives inside the maximum in (6). They remain correlated with the same z in the mass, span and cardinality rows. The common model may include unused integer hole values and real intermediate values; this is a valid relaxation, not an assertion that those values arise from a set. The additional constant cap $\delta \leq C_* - 86j$ is retained to select the row policy. For every actual state the checker verifies membership in the interval, the common side floors, the equality of the scalar and three-row caps, and domination by the group model, with the group's fixed row policy. There are 974,444 row checks over all 3,178 retained states. For example, the group $(a, b, u_p, u_q) = (36, 41, 148, 193)$ has $c \leq 177$ and $z \leq 145$, but also $c + z \leq 231$. The pair of separate maxima $(177, 145)$ violates this last inequality by 91. It is a relaxation artifact, not a feasible set. There are 17 zero-empty, 46 one-empty and 6 two-empty groups, hence

$$1350 \cdot 4 + 1746 \cdot 70 + 82 \cdot 578 = 175016 \quad \longrightarrow \quad 17 \cdot 4 + 46 \cdot 70 + 6 \cdot 578 = 6756$$

high-hole root models. For positive low holes, use the established transport to $z = 14$ after imposing the sharper, z-independent cap.

5 Complete necessary models and conditional rows

We retain the twenty-center, eighteen-unit covering grammar and all actual-contributor choices from [2, 1]. The numbers of modes with $j = 0, 1, 2, 3$ empty units are $4, 70, 578, 2992$. Missing units may not be removed from the root list. In the potentially split pair, either actual member is retained as a separate choice; its positivity inequality is necessary, not sufficient for contribution.

Write $x_0 = R_p < x_1 < x_2 < x_3 < x_4 = M_p$, and similarly y_i on the other side, and put $K = |2D|$. The unchanged mass rows on side p are

$$\begin{aligned} 2M_p + 6R_p - 2S_p + 2z &\le -2T_a - 12 - 2u_p, \\ M_p + 4R_p - S_p &\le -T_{a-1} - 6, \\ x_1 + x_2 + x_3 + M_p - S_p + z &\le -T_a - u_p, \\ R_p + x_1 + x_2 + x_3 + M_p - S_p &\le -T_{a-1}, \end{aligned}$$

with $u_p = 0$ for Z, $u_p = a - 3$ for P, and $u_p = U(a, z)$ (constant in the group) in a high group. The gap rows include $R_p \ge a$, $M_p - R_p \ge 4$ and $x_i - x_{i-1} \ge 1$. Reflection supplies side q. The new global numerical rows are

$$\begin{aligned} \delta &\le 305 - 86j - I && (Z), \\ \delta &\le 242 - 86j - I && (P), \\ R_p + R_q + \delta - 4z &\le 386 - I - 86j, \\ S_p + S_q + \delta - 2K - 4z &\le 1632 - 86j. \end{aligned}$$

For high groups use the three coupled cap rows and retain the same variable z in the span and cardinality inequalities; only the low-hole positive models are transported to $z = 14$.

The row policy is determined from a proved group-wide residual cap C_g, never supplied by the certificate. If $r_{\min}$ is the smallest side size in the group, the conditions are

Rows	Required condition
Ordinary quantiles	$C_g < r_{\min} + 5$
Core quantiles	$C_g < 82$
Basic leading quantiles	$C_g < 171$
Nested leading quantiles	$C_g < 163$
Strict outer order	$j \le 1$, or $j = 2$ with $C_g < 171$

Choose the strongest justified individual quantile type. At a positive center v the global inequalities are $v + l + M_p \ge 86$ and $v - u - M_q \le S_p - 85$; core versions replace $M_p, M_q, 86, 85$ by $R_p, R_q, 82, 81$. Ordinary versions are $v + l \ge b + 5$ and $v - u \le S_p - a - 4$. Individual loss caps are 85 globally and 81 for core rows. In particular, Z_3 now has cap 47 and uses core rows, not the unconditional ordinary rows valid at the preceding endpoint. High two-empty groups have cap at most 12 and satisfy the strict-outer condition. The high one-empty cap can be 98, so core rows are not universally justified there.

Joint block variables represent the actual whole-block covering deficit. For k ordinary centers in a block the valid bound is

$$O \ge 78k - (R_p + R_q + v_{\max} - v_{\min} + 1) + \max_v (l_v + u_v).$$

Let V be the retained own centers in a block. For its leading positive slot and a nonempty subset $S \subseteq V$,

$$O \ge 78(|S| + 2) - 2 - p_{\max} - R_p - 2R_q + \max\{L_p, l_v + u_v : v \in S\} + \sum_{v \in V \setminus S} (l_v + u_v).$$

The analogous sum-of-individual-loss bounds also hold. Each is a lower bound on the same O; the separate gains are not added together. For an own-center ordinary block, its leading-slot loss is added

to the ordinary hull and individual-loss bounds, as in the inherited grouping proof. Only disjoint blocks are summed, with ungrouped losses, to at most δ. These are the proved grouping inequalities of [2], now with $e = 78$. Leading-center quantiles use $2d - 1 = 171$ and $2e + 7 + 2i = 163 + 2i$, respectively, under their stated cap tests.

The coarse model substitutes $K \le M_p + M_q + R_p + R_q + 21$ into the cardinality row and eliminates K. The nested model retains all five interval-envelope levels and the 25 independent-cutoff tail rows

$$K \le M_p + M_q + x_i + y_h + 1 + T_{4-i} + T_{4-h}, \qquad 0 \le i, h \le 4.$$

All models use nonnegative real coordinates as a relaxation of actual configurations. Four proved binary alternatives are available: core quantiles or loss at least 82; ordinary quantiles or an outside endpoint; a loss count at least i or its inner-element cutoff; and both contributors or a sole contributor in the split pair. Both children must be certified. Endpoint alternatives after transport are chosen using the enlarged endpoint, exactly as proved in [1]. No bound on the total number of large-loss centers is imposed.

6 Exact certificate coverage and reproduction

The verifier rebuilds the scalar domain, all mode choices, row policies, and the complete root list before reading certificate assignments. There are $69192 + 6756 = 75948$ integer root systems. The final proof has

Quantity	Count
Structural groups	11,052
Coarse / nested groups	10,958 / 94
Symbolic interval checks	11,090
Integer extremum evaluations	23,917
Integer roots requiring trees	284
Binary internal nodes / leaves	943 / 1,227
Maximum tree depth	12
Final integer leaf systems	76,891
Distinct integer vectors	1,539

For $Ax \le b$, $x \ge 0$, each nonnegative integer certificate vector w satisfies $w^T A \ge 0$ and $w^T b < 0$, giving a strict contradiction. The smallest normalized gap $-w^T b / \sum w_i$ is $1/196$. Symbolic interval checking uses directly constructed quadratic right-hand sides, exact rational endpoints, and the integers around the vertex; it never infers a polynomial from samples. Every integer parameter is assigned once. Binary leaves are checked individually, and omitted children, unknown splits, repeated splits and unused vectors are rejected.

The 23 negative controls include omitted high two-empty modes, the invalid universal core shortcut, arbitrary reflection, an unproved size-16 prefix inverse, a wrong integer inverse, independent hole/cap maxima, and an attempted exclusion of size 93. The conclusion is $N \le 93$, not $N \le 92$.

Reproduction and scope

Only Python 3.10+ and its standard library are required:

```
python -I -S -B verification/nonap_inverse.py
python -I -S -B verification/upper93.py --output .work/upper93.json
python -I -S -B verification/upper93.py --exhaustive
```

The data file is `data/upper93_certificates.json`. The exhaustive mode rebuilds every integer model, checks every row against its symbolic expression, and checks each multiplier pointwise. Here the expanded integer parameter is a. In a high-hole group, z remains a model coordinate: each exact row combination certifies every real value in its interval simultaneously, including every actual integer hole count, rather than testing only the endpoints of that interval. The optional `experiments/generate_upper93_certificates.py` uses SciPy only for discovery; every saved certificate must pass exact checking. Results, the nested-group index, generation logs, report review and input hashes are in `results/2026-10-16/upper93/`.

Standard and optimized Python outputs agree byte for byte. Exhaustive and interval results agree apart from the mode label. Regeneration is byte-identical, and upper-94 regression is unchanged. `verify_upper93_windows.bat` runs all nine finite continuations. The infinite mathematical reductions are proved here and in the cited foundation; program PASS is not a proof-assistant formalization. The next endpoint is cardinality 93 and needs its own lattice and complete coverage checks before any further bound can be asserted.

References

[1] Sum-of-Subset research project, *A sharper non-progression lemma and the audited bound* $N \leq 94$, `reports/upper94_from95_20261016.tex`, 16 October 2026.

[2] Sum-of-Subset research project, *Joint covering losses and the audited bound* $N \leq 95$, `reports/upper95_from96_20261016.tex`, 16 October 2026.

[3] Sum-of-Subset research project, *The audited continuation from 115 to 103*, `reports/upper103_from115_20261014.tex`, 14 October 2026.

[4] Sum-of-Subset research project, *Complete upper-86 research report*, `reports/upper86_complete_20261005.tex`, 5 October 2026. The subsection "Dense spans and the stronger non-AP inverse" motivates Section 2; no archived endpoint exclusion is assumed.